Case Study II: Styrian Health Data

Week 3 Update: Modelling Improvements

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Focus of This Week

- Improved the modelling pipeline based on feedback
- Focused on multi-output Random Forest regression
- Added temporal and environmental features
- Extended analysis with classification and residual diagnostics

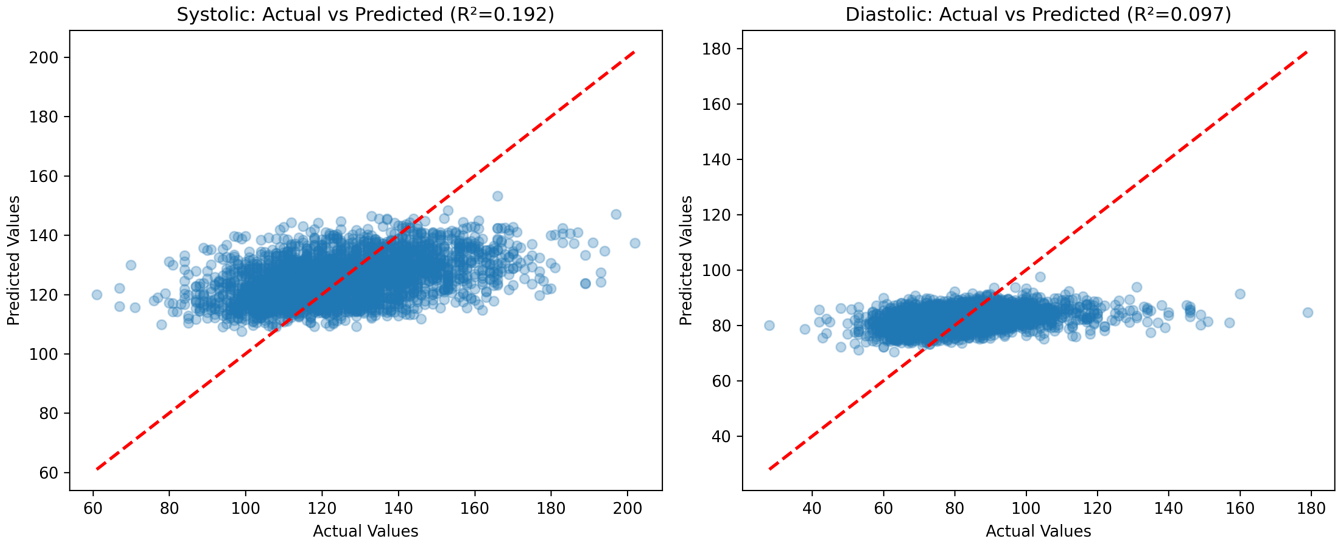
Feedback Addressed

- Avoid relying only on default train-test split
- Clarify feature importance interpretation
- Focus on the multi-output model
- Handle categorical variables more appropriately
- Explore temporal and environmental influences
- Extend analysis beyond basic regression

Updated Modelling Pipeline

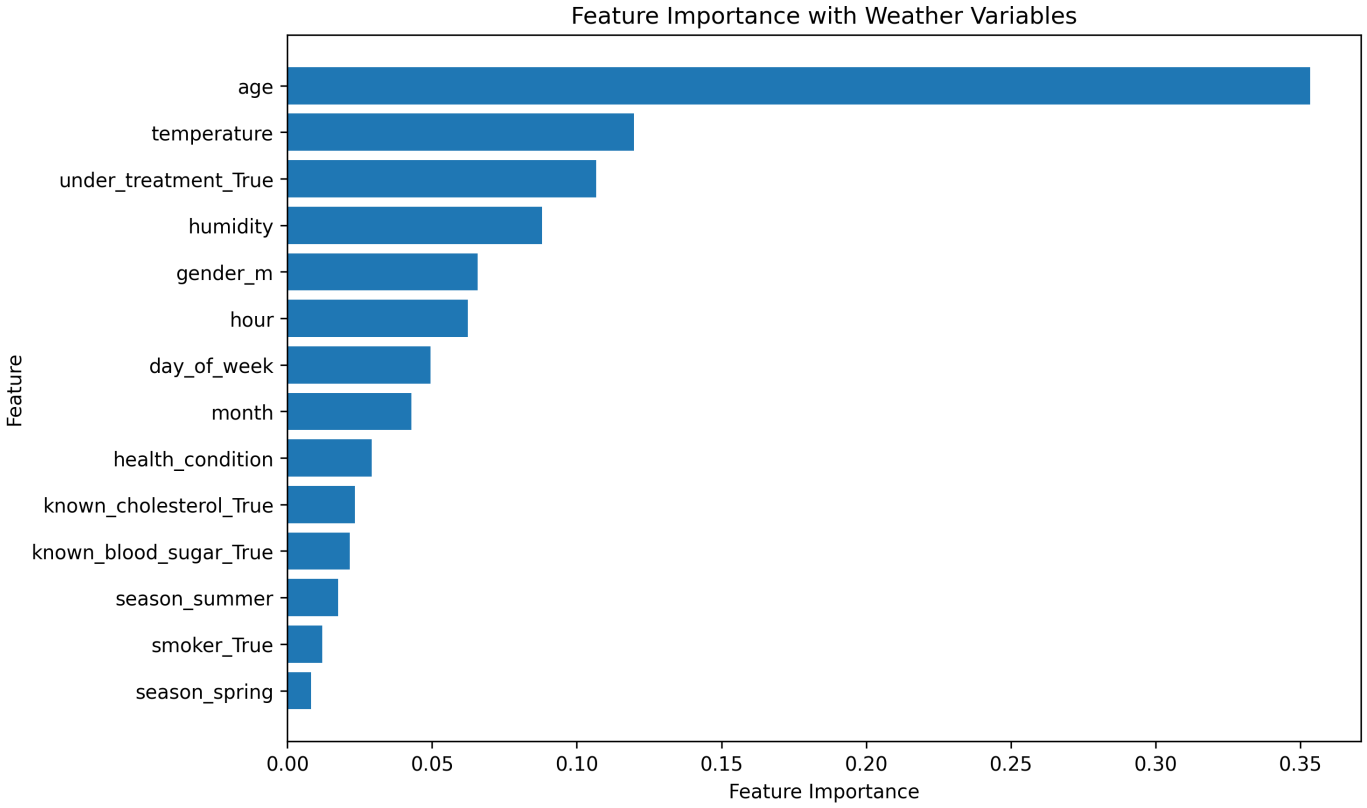
- One-hot encoding for categorical variables
- Stratified train-test split by age group
- Cross-validation for robust evaluation
- Hyperparameter tuning for tree depth
- Temporal feature engineering:
  - hour
  - weekday
  - month
  - season
- Weather integration:
  - temperature
  - humidity

Regression Performance Improvement



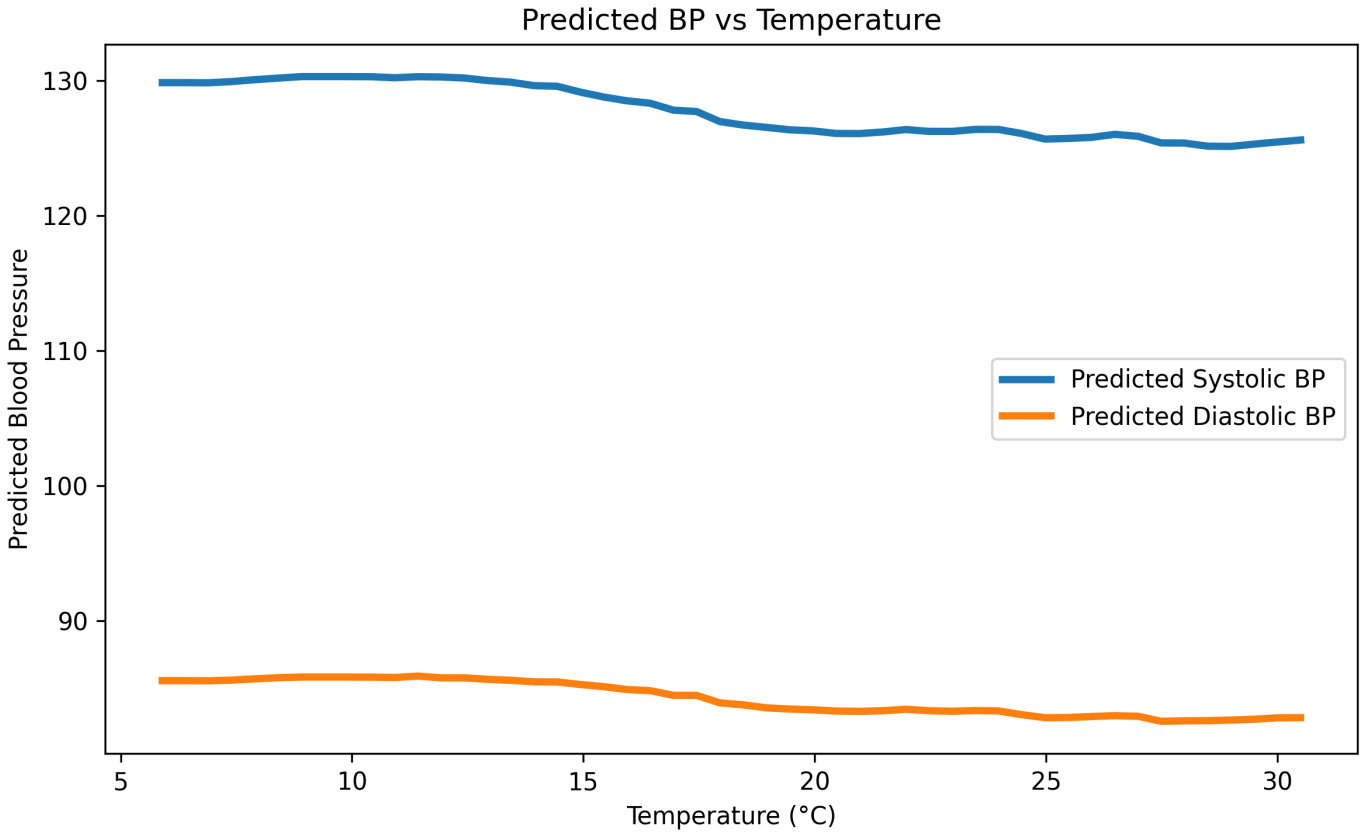
Temporal and environmental features improved predictive performance for both systolic and diastolic BP.

Feature Importance with Weather Variables



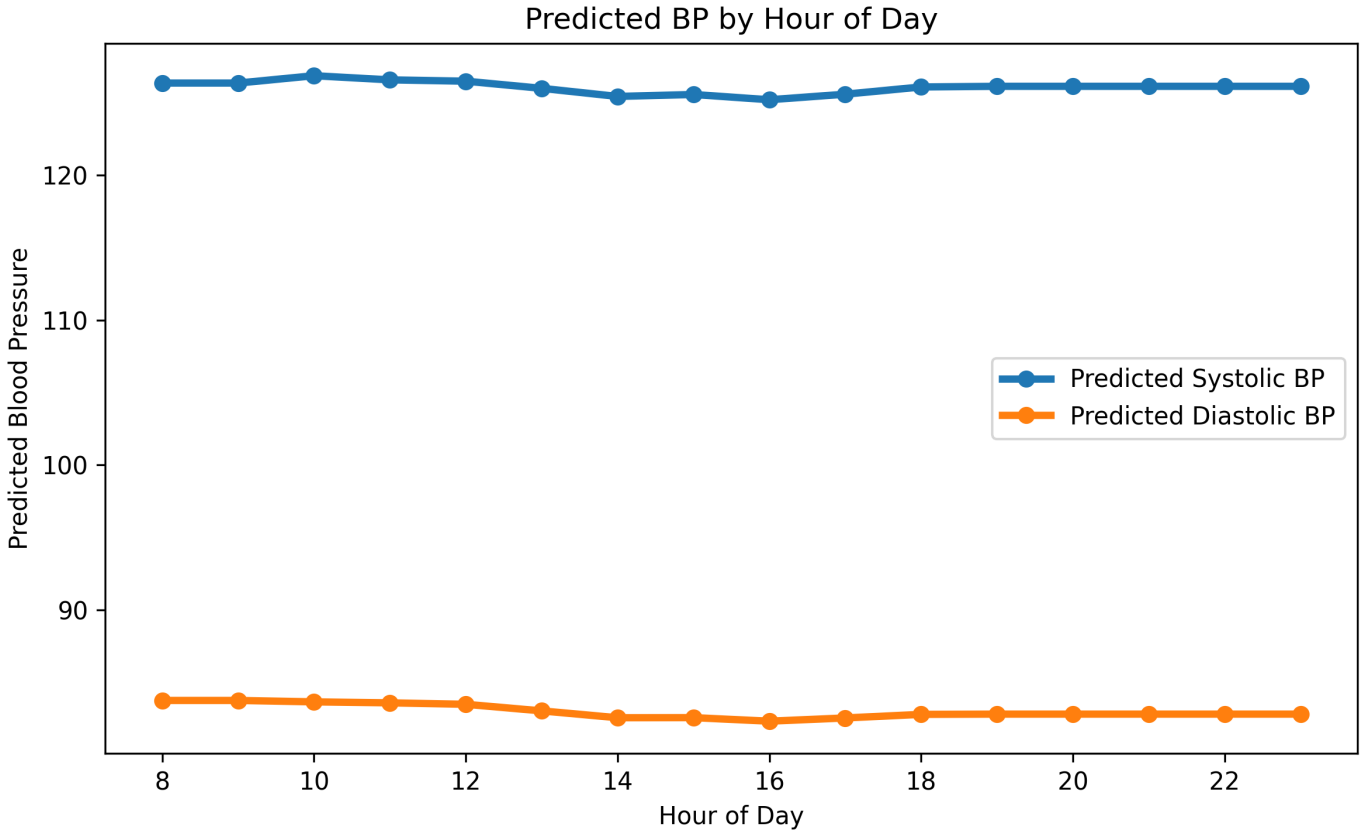
Temporal variables contributed meaningful predictive information alongside age and treatment status.

Predicted Blood Pressure vs Temperature



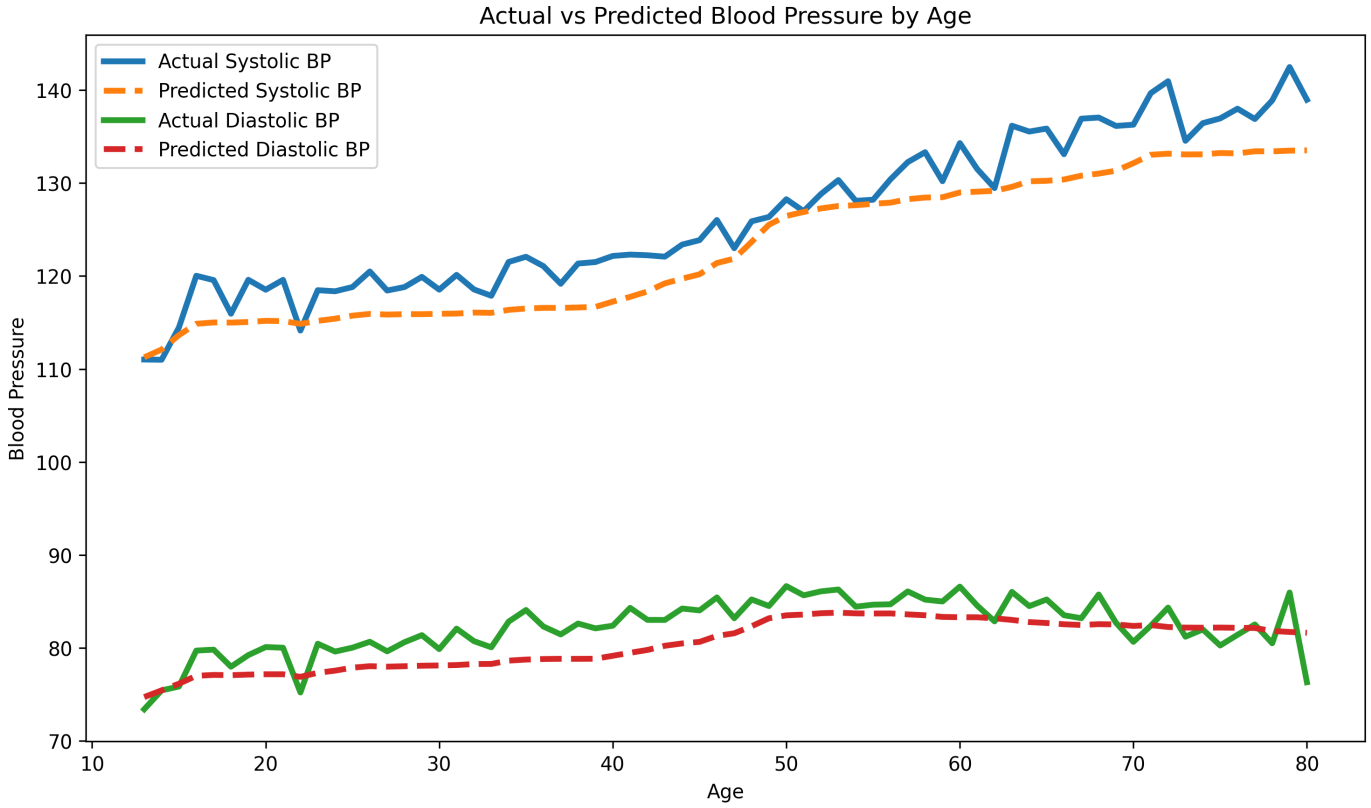
The weather-enhanced model learned a weak inverse relationship between temperature and predicted BP.

Predicted Blood Pressure by Hour



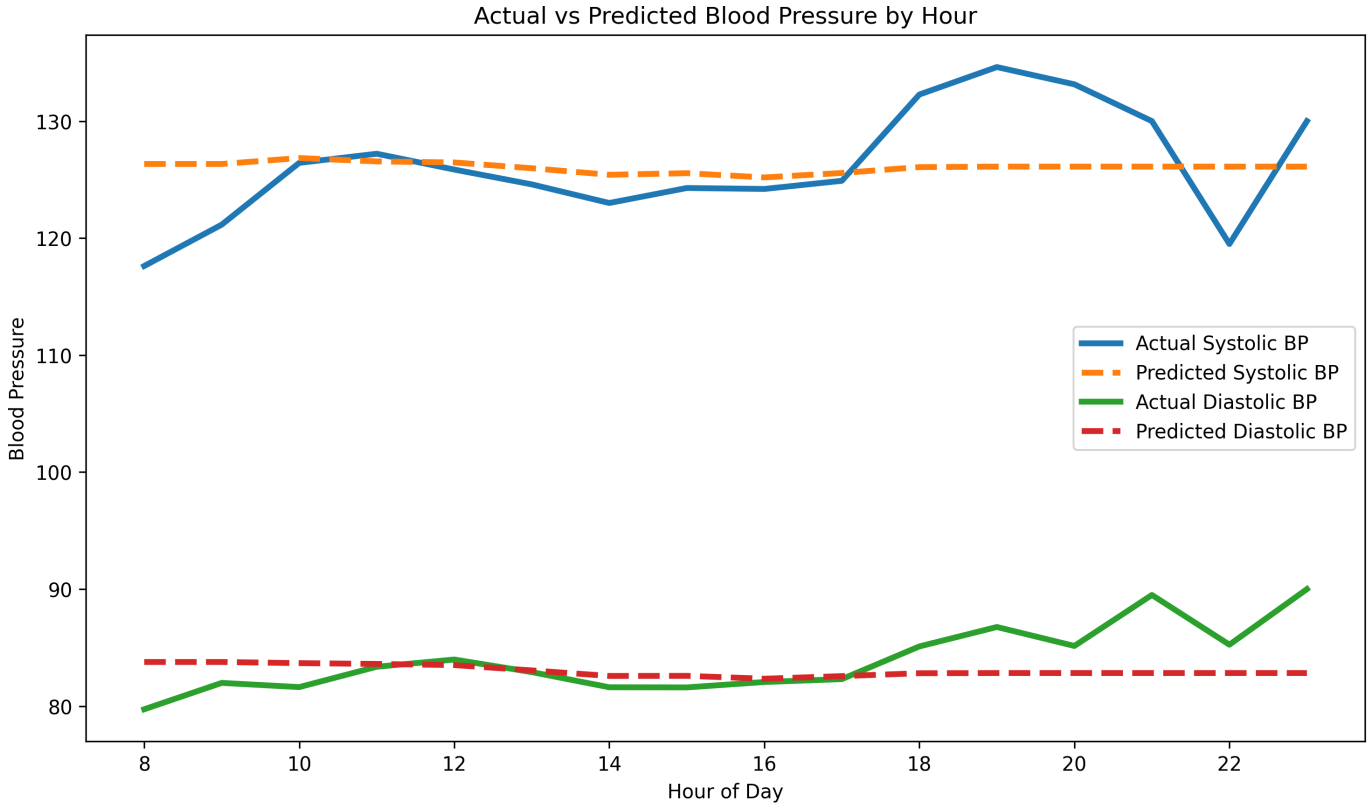
The model captured modest within-day temporal variation.

Actual vs Predicted BP by Age



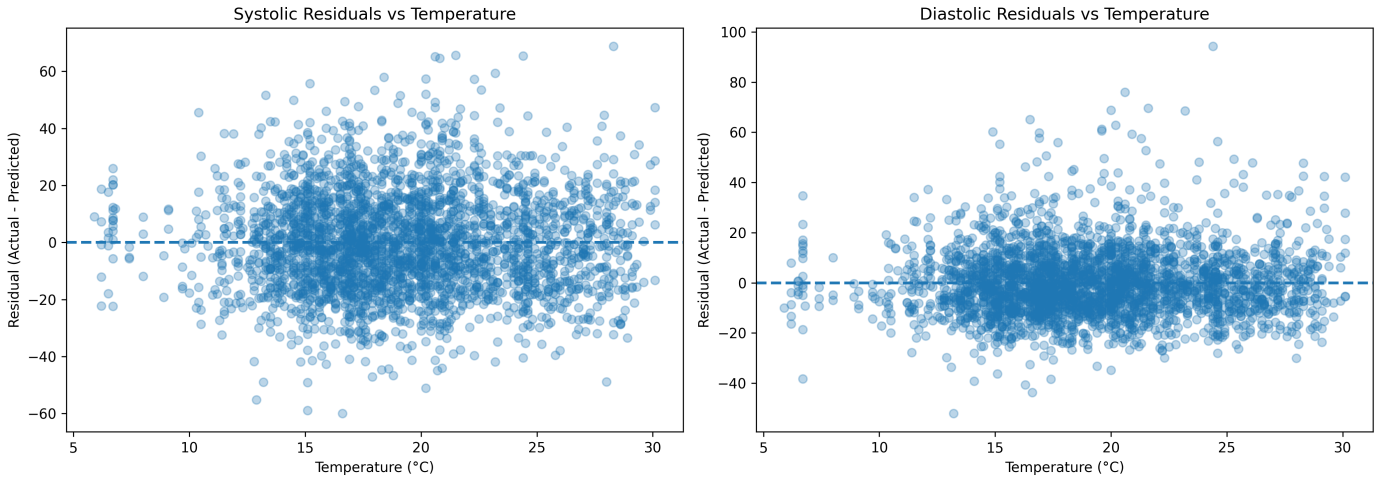
The model captured broader age-related BP trends despite noisy individual predictions.

Actual vs Predicted BP by Hour



The model captured broad temporal structure but struggled with finer hourly fluctuations.

Residual Diagnostics



Residuals remained centered across temperature ranges, suggesting reduced systematic environmental bias.

Classification Extension

- Regression predicts continuous BP values
- Classification predicts elevated BP categories
- Systolic high BP:

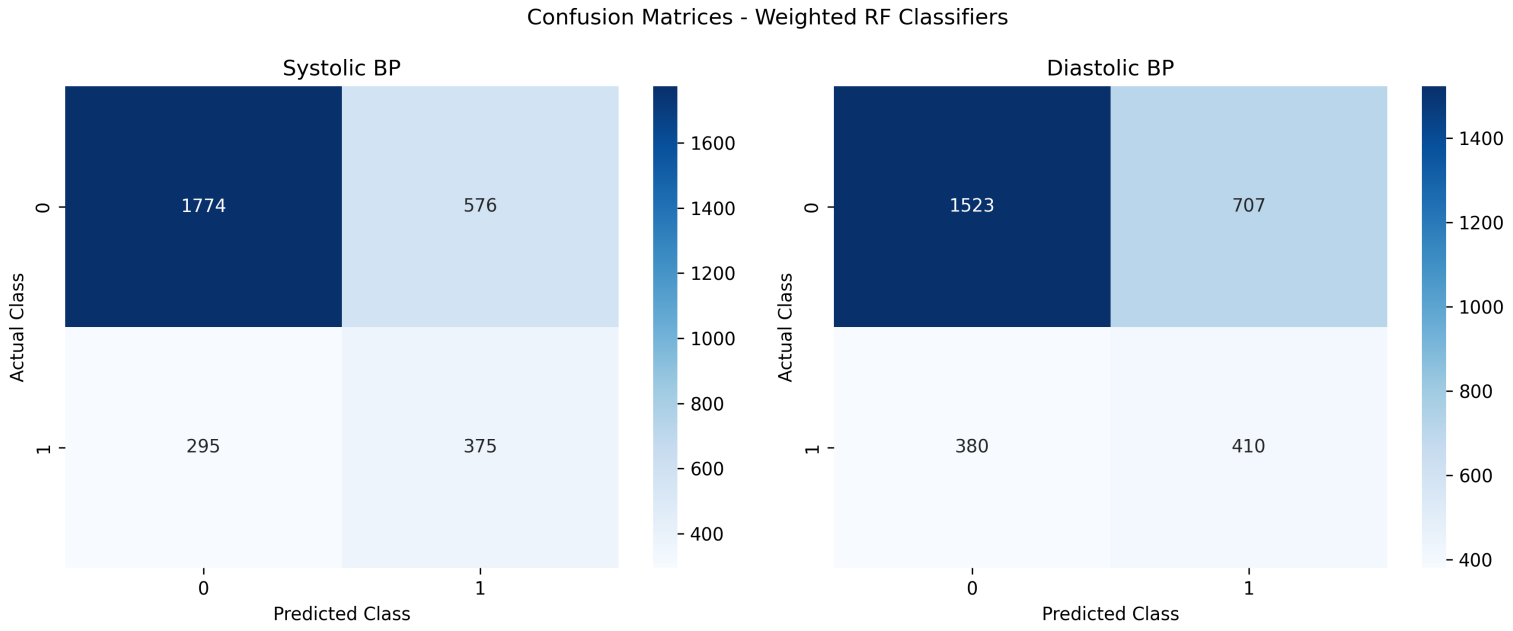
$$BP_{sys} \geq 140$$

- Diastolic high BP:

$$BP_{dia} \geq 90$$

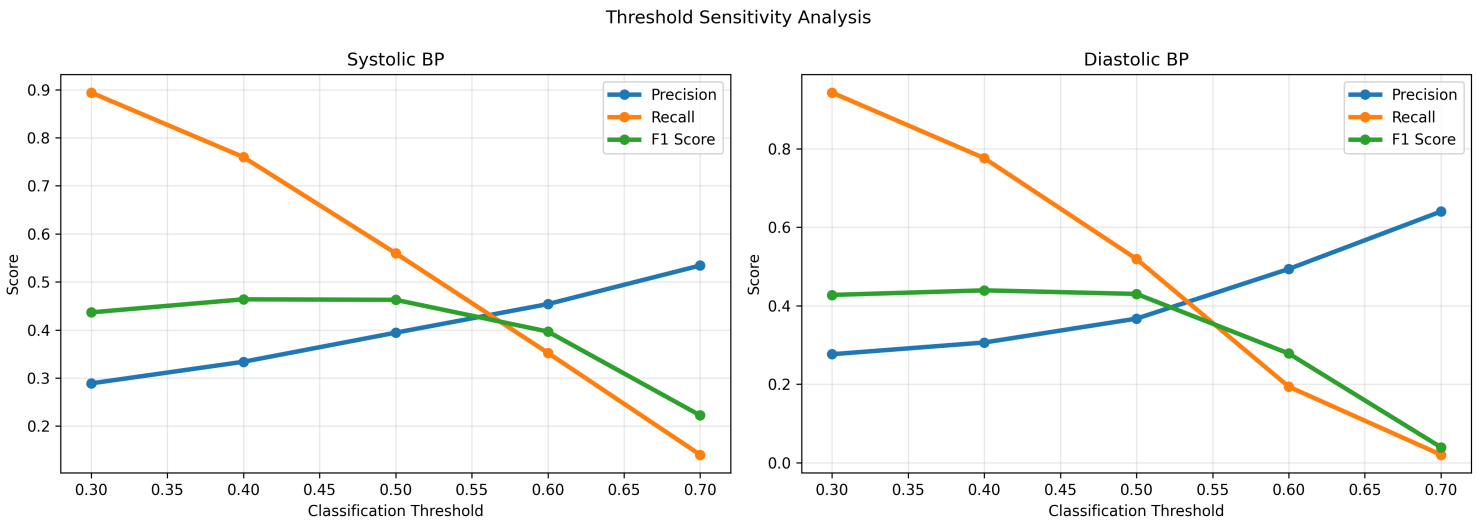
- Weighted Random Forest used to handle imbalance

Classification Confusion Matrices



Systolic BP classification performed slightly better than diastolic classification.

Threshold Sensitivity Analysis



Lower thresholds improved recall while higher thresholds improved precision.

Current Takeaways

- Temporal and weather features improved regression performance
- The model captured population-level trends better than individual variation
- Weighted classification improved elevated BP detection
- Threshold tuning demonstrated the precision–recall tradeoff
- Residual analysis revealed remaining prediction variability